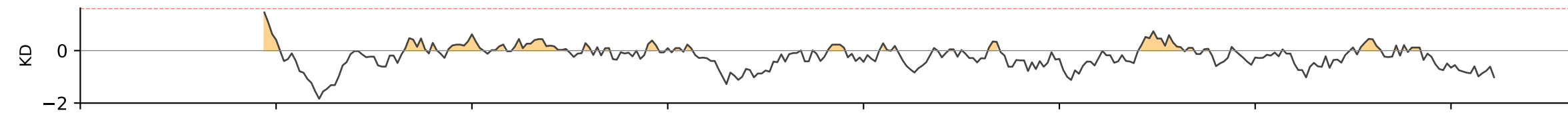
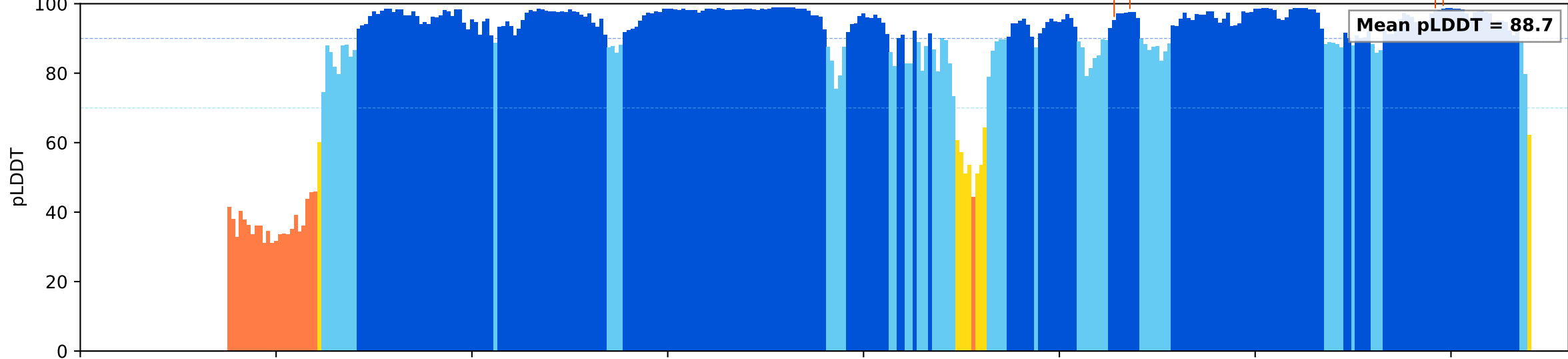


PL25_M23 — bifunctional cell-wall hydrolase architecture (Slt35-/Lysozyme-like fold + Pfam Peptidase_M23, CwIP-like)



N-term
hydrophobic
region

Slt35-/Lysozyme-like fold
(CDD cd13399; SSF53955; Gene3D)
GLYCAN HYDROLASE — fold-level assignment

linker

M23 peptidase
(Pfam PF01551, predicted Zn²⁺-binding)
PUTATIVE PEPTIDE HYDROLASE

Putative Zn²⁺-binding residues (canonical HxxxD + HxH; H264, D268, H346, H348)

Residue position in full-length PL25_M23 (aa)

Residue numbering: full-length PL25_M23 (370 aa). The N-terminal hydrophobic region (residues 19-46, orange band) overlaps the start of the cleavable signal peptide. The N-terminal glycoside-hydrolase domain is detected at the FOLD level by CDD cd13399 ($E = 1.1 \times 10^{-27}$), SUPERFAM, and Gene3D. The Pfam Peptidase_M23 domain is detected at the Pfam level by Pfam PF01551 ($E = 1.1 \times 10^{-15}$). The CwIP-like architecture is inferred from sequence motifs and AlphaFold side-chain geometry only; experimental confirmation (ED/MS/MS assay) remains to be performed.

- Slt35-/Lysozyme-like glycoside hydrolase (fold-level)
- N-terminal hydrophobic region (residues 19-46) (no cleavable signal peptide)
- Putative M23 metallopeptidase (Pfam PF01551)
- Putative Zn²⁺-binding residues
- CwIP-like architecture